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Bihar Engineering University, Patna

B.Tech 2nd Semester Exam., 2019

Subject: MATHEMATICS-II (Ordinary Differential Equations and Complex Variables)

Code: 102202 | ***Time: 3 hours*** | ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Answer the following (any seven): **[2×7=14]**

(a) Find the directional derivative of $\varphi(x,y,z)=x^2yz+4\sim xz^2$ at $(1,-2,-1)$ in the direction $2i-j-2k$

(b) Evaluate $\nabla.[r\nabla(1/r^3)]$.

(c) What is the degree of the differential equation $(d^3y/dx^3)^{2/3}+(d^3y/dx^3)^{3/2}=0$?

(d) Find the general solution of the differential equation $x(x^2+3y^2)dx+y(y^2+3x^2)dy=0$

(e) Evaluate the integral $\int_C[(e^z+\sin\sim\pi z)dz] / [(z-1)(z+1)(z+4)]$, C:$|z|=2$

(f) Evaluate the integral $\int_C[dz] / [(z^2+4z+3)^2]$ C:$|z|=4$

(g) Define the pole-type singularity with an example.

(h) Find the bilinear transformation that maps $z_1=\infty$, $z_2=i$ and $z_3=0$ into the points $w_1=0$, $w_2=i$ and $w_3=\infty$.

(i) If $a<b$, then evaluate the integral $\int_a^b|(x-a)+(x-b)|dx$

(j) Evaluate the integral $\int_0^\infty\int_x^\infty[e^{-y}] / [y]dy\sim dx$

Q.2 Solve both questions: **[14]**

(a) Evaluate the integral $\int_0^a\int_y^a[x] / [(x^2+y)^2]dy\sim dx$

(b) Find the mass of a plate in the form of a quadrant of an ellipse $x^2/a^2+y^2/b^2=1$ whose density per unit area is given by $\rho=kxy$.

Q.3 Solve both questions: **[14]**

(a) Evaluate $\int_C F\cdot dr$, where $F=(3x^2+6y)i-14yzj+20xz^2k$ from (0,0,0) to (1,1,1) along the path $x=t$, $y=t^2$ and $z=t^3$

(b) Evaluate $\int_C F\cdot dr$ along the straight line joining (0,0,0) to (1,1,1)

Q.4 Solve both questions: **[14]**

(a) Solve the differential equation $(x^2+y^2+x)dx-(2x^2+2y^2-y)dy=0$

(b) Solve the differential equation $y=2px+y^2p^3$

Q.5 Solve both questions: **[14]**

(a) State and prove Rodrigues' formula.

(b) Show that $2nJ_n(x)=x[J_{n+1}(x)+J_{n-1}(x)]$

Q.6 Solve this question: **[14]**

Find the series solution of the differential equation $x^2d^2y/dx^2+6\sim xdy/dx+(x^2+6)y=0$

Q.7 Solve both questions: **[14]**

(a) State and prove the sufficient condition for a function $w=f(z)$ to be analytic.

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(b) Find an analytic function $f(z)$ such that $\mathrm{Re}\{f^{\mathrm{prime}}(z)\}=3x^2-4y-3y^2$ and $f(1+i)=0$

Q.8 Solve both questions: **[14]**

(a) Discuss the nature of the singularities for $([1-\cosh\sim z] / [z^3])$. Also determine the order of the pole and corresponding residue if it exists.

(b) Find what regions of the w-plane correspond by the transformation $w=([z-i] / [z+i])$ to the interior of a circle of centre $z=-i$.

Q.9 Solve both questions: **[14]**

(a) Evaluate $\mathrm{oint}_C[\sin^2 z] / [z(z-1)(2z+5)]dz$, C:$|z-1|+|z+1|=3$

(b) Evaluate $\int_0^{\infty}[\sin(mx)] / [x(x^2+a^2)]dx$

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